

SQL Joins — Lab Report

Google Cybersecurity Certificate
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Project Description

This lab activity focuses on using SQL JOIN operations within a MariaDB relational database to support a simulated security incident investigation. In the scenario, several machines within an organisation have been compromised, and the task is to retrieve relevant records from the database to aid the investigation. The organisation database contains multiple tables — specifically **machines**, **employees**, and **log_in_attempts** — each holding different aspects of the organisation's infrastructure and activity data.

The lab required performing three types of JOIN operations: an INNER JOIN to match employees to their assigned machines, a LEFT JOIN and a RIGHT JOIN to identify unmatched records across both tables, and finally an INNER JOIN to correlate employees with their login activity. Each task demonstrates how relational databases use a shared column (a foreign key) to link related tables, enabling security analysts to draw meaningful conclusions from distributed data.

Tasks

Task 1 — Match Employees to Their Machines (INNER JOIN)

The first task required identifying which employee is using which machine. Both the **machines** and **employees** tables share the **device_id** column, making it the natural join key. An INNER JOIN returns only the rows where a match exists in both tables — meaning only machines that are assigned to a known employee (and vice versa) appear in the result.

The query below selects all columns from both tables and joins them on their shared **device_id** column. Machines with no assigned employee and employees with no machine are excluded from the output.

```
SELECT *  
FROM machines  
INNER JOIN employees ON machines.device_id = employees.device_id;
```

```

MariaDB [organization]> SELECT *
->
-> FROM machines
->
-> INNER JOIN employees ON machines.device_id = employees.device_id;

```

device_id	operating_system	email_client	OS_patch_date	employee_id	employee_id	device_id	username	department
a320b137c219	OS 2	Email Client 2	2021-03-01	1000	1000	a320b137c219	el Larson	Marketing
East-170								
b239c825d303	OS 1	Email Client 1	2021-03-01	1001	1001	b239c825d303	bmoreno	Marketing
Central-276								
c116d593e558	OS 3	Email Client 1	2021-09-01	1002	1002	c116d593e558	tshah	Human Resources
North-434								
d394e816f943	OS 3	Email Client 2	2021-03-01	1003	1003	d394e816f943	sgillmore	Finance
South-153								
e218f877g788	OS 2	Email Client 1	2021-09-01	1004	1004	e218f877g788	eraab	Human Resources
South-127								
f551g340h864	OS 3	Email Client 2	2021-12-01	1005	1005	f551g340h864	gesparza	Human Resources
South-366								
g329h357i597	OS 1	Email Client 2	2021-06-01	1006	1006	g329h357i597	alevitsk	Information Technology
East-320								
h174i497j413	OS 2	Email Client 1	2021-03-01	1007	1007	h174i497j413	wjaffrey	Finance
North-406								
i858j583k571	OS 2	Email Client 2	2021-06-01	1008	1008	i858j583k571	abernard	Finance
South-170								
k242l212m542	OS 1	Email Client 1	2021-03-01	1010	1010	k242l212m542	jlansky	Finance
South-109								
l748m120n401	OS 3	Email Client 1	2021-09-01	1011	1011	l748m120n401	drosas	Sales
South-292								
m756n668o146	OS 1	Email Client 2	2021-09-01	1012	1012	m756n668o146	nmason	Information Technology
North-160								

Figure 1 — INNER JOIN result: machines table joined to employees table on device_id. The query returned 185 rows, each representing a matched employee-machine pair.

Result: The INNER JOIN returned **185 rows**. Each row represents a machine that is definitively assigned to an employee in the organisation, providing a clean mapping for the incident investigation.

Task 2 — Return Unmatched Records (LEFT JOIN & RIGHT JOIN)

The second task extended the investigation to unmatched records. This required two separate queries: a LEFT JOIN to retrieve all machines (including those with no assigned user), and a RIGHT JOIN to retrieve all employees (including those with no machine assigned).

LEFT JOIN

A LEFT JOIN returns every record from the left table (**machines**) regardless of whether a matching record exists in the right table (**employees**). Where no match is found, the employee columns are filled with NULL.

```

SELECT *
FROM machines
LEFT JOIN employees ON machines.device_id = employees.device_id;

```

```

MariaDB [organization]> SELECT *
->
-> FROM machines
->
-> left JOIN employees ON machines.device_id = employees.device_id;

```

device_id	operating_system	email_client	OS_patch_date	employee_id	employee_id	device_id	username	department
office								
a320b137c219	OS 2	Email Client 2	2021-03-01	1000	1000	a320b137c219	elarson	Marketing
East-170								
b239c825d303	OS 1	Email Client 1	2021-03-01	1001	1001	b239c825d303	bmoreno	Marketing
Central-276								
c116d593e558	OS 3	Email Client 1	2021-09-01	1002	1002	c116d593e558	tshah	Human Resources
North-434								
d394e816f943	OS 3	Email Client 2	2021-03-01	1003	1003	d394e816f943	sgillmore	Finance
South-153								
e218f877g788	OS 2	Email Client 1	2021-09-01	1004	1004	e218f877g788	eraab	Human Resources
South-127								
f551g340h864	OS 3	Email Client 2	2021-12-01	1005	1005	f551g340h864	gesparza	Human Resources
South-366								
g329h357i597	OS 1	Email Client 2	2021-06-01	1006	1006	g329h357i597	alevitsk	Information Technology
East-320								
h174i497j413	OS 2	Email Client 1	2021-03-01	1007	1007	h174i497j413	wjaffrey	Finance
North-406								
i858j583k571	OS 2	Email Client 2	2021-06-01	1008	1008	i858j583k571	abernard	Finance
South-170								
k242l212m542	OS 1	Email Client 1	2021-03-01	1010	1010	k242l212m542	jlansky	Finance
South-109								
l748m120n401	OS 3	Email Client 1	2021-09-01	1011	1011	l748m120n401	drosas	Sales
South-292								
m756n668o146	OS 1	Email Client 2	2021-09-01	1012	1012	m756n668o146	nmason	Information Technology

Figure 2 — LEFT JOIN result: all machines returned, with NULL in employee columns where no match exists. The last record's username column shows NULL.

Result: The LEFT JOIN returned all machines from the **machines** table. The final record's **username** column contained **NULL**, indicating at least one machine has no employee assigned to it — a potential security flag.

RIGHT JOIN

A RIGHT JOIN returns every record from the right table (**employees**) regardless of whether a matching machine record exists. Where no match is found, the machine columns are filled with NULL. This reveals employees who have not been assigned any machine.

```

SELECT *
FROM machines
RIGHT JOIN employees ON machines.device_id = employees.device_id;

```

device_id	operating_system	email_client	OS_patch_date	employee_id	employee_id	device_id	username	department
office								
a320b137c219	OS 2	Email Client 2	2021-03-01	1000	1000	a320b137c219	el Larson	Marketing
East-170								
b239c825d303	OS 1	Email Client 1	2021-03-01	1001	1001	b239c825d303	bmoreno	Marketing
Central-276								
c116d593e558	OS 3	Email Client 1	2021-09-01	1002	1002	c116d593e558	tshah	Human Resources
North-434								
d394e816f943	OS 3	Email Client 2	2021-03-01	1003	1003	d394e816f943	sgillmore	Finance
South-153								
e218f877g788	OS 2	Email Client 1	2021-09-01	1004	1004	e218f877g788	eraab	Human Resources
South-127								
f551g340h864	OS 3	Email Client 2	2021-12-01	1005	1005	f551g340h864	gesparza	Human Resources
South-366								
g329h357i597	OS 1	Email Client 2	2021-06-01	1006	1006	g329h357i597	alevitsk	Information Technology
East-320								
h174i497j413	OS 2	Email Client 1	2021-03-01	1007	1007	h174i497j413	wjaffrey	Finance
North-406								
i858j583k571	OS 2	Email Client 2	2021-06-01	1008	1008	i858j583k571	abernard	Finance
South-170								
NULL	NULL	NULL	NULL	NULL	1009	NULL	lrodriqu	Sales
South-134								
k242l212m542	OS 1	Email Client 1	2021-03-01	1010	1010	k242l212m542	jlansky	Finance
South-109								
l748m120n401	OS 3	Email Client 1	2021-09-01	1011	1011	l748m120n401	drosas	Sales

Figure 3 — RIGHT JOIN result: all employees returned. Row with employee_id 1009 (Irodriqu) shows NULL for all machine columns, indicating no machine is assigned to that employee.

Result: The RIGHT JOIN revealed at least one employee (Irodriqu, employee_id 1009) with no machine assigned — all machine-related columns returned NULL for that record. The final record's **username** column contained **areyes**.

Task 3 — Retrieve Login Attempt Data (INNER JOIN)

The third task correlated employee records with login activity. The **employees** table and the **log_in_attempts** table share the **username** column. An INNER JOIN on this column returns only the login events where the username can be matched to a known employee — filtering out any orphaned or unrecognised login attempts.

The resulting dataset combines employee metadata (department, office, device) with login event details (event_id, login_date, login_time, country, IP address, success flag), giving the analyst a full picture of each authenticated session.

```
SELECT *
FROM employees
INNER JOIN log_in_attempts ON employees.username = log_in_attempts.username;
```

```

MariaDB [organization]> select *
-> from employees
-> inner join log_in_attempts on employees.username = log_in_attempts.username;

```

employee_id	device_id	username	department	office	event_id	login_date	login_time	country	ip_address
1032	g773h303i639	jrafael	Information Technology	Central-309	1	2022-05-09	04:56:27	CAN	192.168.243.140
1026	a998b568c863	apatel	Human Resources	West-320	2	2022-05-10	20:27:27	CAN	192.168.205.12
1031	f419g188h578	dkot	Marketing	West-408	3	2022-05-09	06:47:41	USA	192.168.151.162
1031	f419g188h578	dkot	Marketing	West-408	4	2022-05-08	02:00:39	USA	192.168.178.71
1032	g773h303i639	jrafael	Information Technology	Central-309	5	2022-05-11	03:05:59	CANADA	192.168.86.232
1020	u899v381w363	arutley	Marketing	South-351	6	2022-05-12	17:00:59	MEXICO	192.168.3.24
1004	e218f877g788	eraab	Human Resources	South-127	7	2022-05-11	01:45:14	CAN	192.168.170.243
1035	j236k303l245	bisles	Sales	South-171	8	2022-05-08	01:30:17	US	192.168.119.173
1033	NULL	yappiah	Information Technology	West-387	9	2022-05-11	13:47:29	MEX	192.168.59.136
1032	g773h303i639	jrafael	Information Technology	Central-309	10	2022-05-12	09:33:19	CANADA	192.168.228.221
1003	d394e816f943	sgilmore	Finance	South-153	11	2022-05-11	10:16:29	CANADA	192.168.140.81
1031	f419g188h578	dkot	Marketing	West-408	12	2022-05-08	09:11:34	USA	192.168.100.158
1027	b806c503d354	mrhah	Marketing	West-246	13	2022-05-11	09:29:34	USA	192.168.100.158

Figure 4 — INNER JOIN on employees and log_in_attempts via username. The query returned 200 records, each pairing an employee profile with a login event.

Result: The INNER JOIN returned **200 records**. Each record pairs an employee's profile information with a corresponding login attempt, enabling the investigation to narrow down which staff members were active during the period and from which locations, countries, and IP addresses — critical context for tracing a security incident.

Summary

This lab demonstrated how SQL JOIN operations are an essential tool in a security analyst's workflow. By joining the **machines**, **employees**, and **log_in_attempts** tables across shared columns, it was possible to construct a coherent picture of the organisation's asset assignments and authentication activity from data that would otherwise be siloed.

Task	Join Type	Tables Joined	Join Column	Records Returned
Task 1	INNER JOIN	machines + employees	device_id	185
Task 2a	LEFT JOIN	machines + employees	device_id	All machines
Task 2b	RIGHT JOIN	machines + employees	device_id	All employees
Task 3	INNER JOIN	employees + log_in_attempts	username	200

Key takeaways from this lab include: INNER JOIN is the strictest match — it surfaces only records that exist in both tables, making it ideal for confirmed asset-employee pairings and authenticated sessions. LEFT JOIN and RIGHT JOIN expose gaps — NULLs in the result signal unassigned machines or employees without assets, both of which can represent security gaps or anomalies worth investigating. Using table-qualified column references (e.g., *machines.device_id*) is essential when joining tables that share column names, to avoid ambiguity errors. Together, these three JOIN types form the foundation for cross-table forensic analysis in security investigations.